More Than Just Attention: How a Hybrid AI Model Outperforms Traditional Volatility Forecasting Methods

Abstract: COVID-19 has created an environment of more volatility in the stock markets, which indicates the shortcomings of traditional methods such as GARCH, thereby necessitating sophisticated hybrid models capable of handling crisis environments. This study has been carried out to understand the superiority of the Convolutional Neural Network-LSTM model in forecasting the volatility in stock exchanges relative to traditional models by demonstrating conclusive empirical evidence. To evaluate the superiority of the model, this study uses the hybrid model to forecast stock exchange by using four major stock exchanges: S&P 500 (USA), NASDAQ (USA technology stocks), NIFTY 50 (India), and IDX (Indonesia). These four stock exchanges represent two emerging economies (India and Indonesia) plus one advanced economy (USA), and the data span from January 1, 2018 to December 31, 2024. The results from our analysis clearly show that although the volatility in the emerging markets is higher than in the advanced economy, in our analysis, it has been clearly established that employing attention mechanisms in forecasting can accurately detect significant factors affecting instability in the markets, precisely during the Covid-19 crisis. Given the above findings, it is highly recommended that investors and policymakers from both developed and developing markets take into consideration highly sophisticated predictive models that have been endowed with attention mechanisms. It is argued that such high-end tools result in better volatility forecasts as an anticipatory measure, thus making decision-making processes more informed and making more stable portfolios in an increasingly AI-driven economic.

Keywords: Volatility forecasting**,** stock market, Covid-19, CNN-LSTM models

1. Introduction

In the contemporary economic landscape, globalization, markets, and especially finance play an important and dominating position in the development and progress of the national and world economies through the finance and investment in the real sector. However, despite their strategic position and significant contribution to the national and world economies, these markets are extremely vulnerable to various disturbances happening across the globe (Masoud, 2013; Yang, 2025). However, over the last ten years, various occurrences, including the Covid-19 pandemic, have exposed this vulnerability as these occurrences have caused considerable market volatilities globally due to geo-political tensions and raw materials, as well as changes in investment portfolios (e.g., Xu et al., 2024; Ye et al., 2025; Hussain & Islam, 2025). All this, therefore, underlines the situation exemplified by the presence of volatilities that exist in the stock exchange, a situation that can be considered serious.

Volatility plays an important role in financial markets and is considered an essential element that predicts the sentiment of investors. It may even emerge to be one of the determinants of risk;

knowing the volatility and forecasting it holds significant importance in taking decisions by investors. Volatility refers to the capacity of asset prices to fluctuate and is a symbols of uncertainty, risk, and possible return. The process of computation of the value of volatility becomes essential while it is being applied to options pricing models; it has also shown great importance in risk management and market efficiency measurements. The ever-rising complexities in financial markets alongside changing trends in financial volatilities along with variable sensitivities resulting from various crises or shocks that have arisen from recent pandemics like Covid-19 have compelled nations to create effective models of forecasts (Kim & Won, 2018; Chen et al., 2022; Xu et al., 2024).

Nevertheless, when financial volatility is being projected, immense challenges remain given its nature, associated with significant levels of noise and switches usually encountered in financial series. Econometric models, even when aesthetically tractable, cannot easily accommodate the dynamism and complexity of the modern financial system. The difficulty of capturing long-run dependence as well as time-varying market conditions by traditional models built upon the assumptions of stationarity processes and linear dependencies cannot be overemphasized, especially in the processes of complicated issues in the shock of macroeconomic factors, sectorial diversities, and investor anomalies (Amirshahi & Lahmiri, 2023; Hu et al., 2023; Adesina & Obokoh, 2025).

Despite classical models, such as GARCH and ARIMA, dominating the current literature in terms of already wide application, they heavily depend upon linear assumptions, failing to account for any latent features that might exist in a dataset, as well as deep non-linear patterns, due to a lack of ability in generalizing, thus rendering them inoperable in times of market turbulences, where volatility is a standard scenario (Kim & Won, 2018; Koo & Kim, 2022). They do not yield much adaptability in ever-changing financial scenarios because they must depend heavily on handcrafted features. The introduction of a deep learning paradigm exemplified by architectures such as recurrent and convolutional deep learning architectures, effectively capture complex relationships through data-driven representation learning (Yu & Li, 2018; Christensen et al., 2023).

While many studies have addressed deep learning (DL) models for price prediction, very few have ventured into volatility forecasting using hybrid architectures. Among those, most of them examine pre-pandemic datasets or are designed for stable markets, hence leaving a window of opportunity for performance evaluation under post-COVID, high-volatility, and emerging markets scenarios. Second, existing CNN-LSTM approaches are mostly geared toward price direction or classification problems rather than being able to provide a fine-grained continuous volatility forecast (Rostamian & O'Hara, 2022; Song et al., 2023). Another uncharted area relates to integrated attention mechanisms or custom-built volatility features within hybrid models.

Therefore, this study puts forth a new hybrid model derived from CNN and LSTM combined with an attention mechanism to bring about a better volatility forecast. CNN is designed to extract spatially aware patterns from multivariate time series containing technical indicators, while LSTM handles the temporal dependencies present across sequences. The attention layer then aids in discerning the importance of certain time steps and indicators at specific moments. In this pipeline, volatility is introduced as a feature to create both learning stability and sensitivity in the system.

Moreover, the model is tested across various market segments, including emerging markets and the post-COVID market. These tests validated the model in diverse financial contexts (Singh et al., 2023; Mohamed Rida et al., 2024). This study also enriches existing research on stock market volatility through a comparative analysis of the four major stock market indices, before and after the COVID-19 crisis, in different countries. Two of these indices are from developed countries (the S&P 500 and NASDAQ indices of the United States), while two of them are from emerging Asian markets (the NIFTY 50 index of India and Jakarta Composite Index (IDX) of Indonesia). Incorporating this geographical and structural heterogeneity allows for a more comprehensive analysis of stock market volatility in the face of recent shocks, such as the pandemic and global events. Moreover, the study emphasizes the practical implications of intelligent econometric prediction models; therefore, CNN and LSTM models are also presented.

While selecting the markets for the study, two broad and prominent markets out of the developed economies (S&P 500 and NASDAQ) are considered. Furthermore, two markets are considered from the emerging or developing countries: NIFTY 50 and IDX. The reasons for choosing the markets are based on their economic significance. The United States is the biggest capital market in the world. We have considered the markets S&P 500 and NASDAQ. India and Indonesia have been chosen as they are two of the biggest emerging economies in Asia with high levels of activity. They share strong economic ties with the United States. Their economies also reflect contrasting characteristics; for example, India's economy is dominated by services as opposed to Indonesia's commodity-driven stock exchange. This selection of indices would provide a holistic review of global sentiments with sectoral and emerging-market representations.

The significance of this study to existing theory is three-fold. Firstly, it adds to the existing literature through a cross-markets/cross-regime analysis of VForecasts during developed and emerging markets in those uncertain periods of Covid-19 and Russia–Ukraine conflicts. Secondly, it expands upon existing models through the addition of engineered hybrid models that integrate CNN, LSTM, and attention to improve VSpikes prediction findings. Thirdly, it adds to existing theory through extensive tests of its models to create increased pragmatic implications for uncertain periods.

In this paper, the structure is as follows: In section 2, an extensive review of the literature for traditional models, LSTM, CNN, and hybrid deep learning outlines for volatility forecast is presented. Section 3 specifically presents the dataset and the technical indicators utilized, just like the feature engineering techniques. On the other hand, section 4 outlines the suggested network architecture of CNN, LSTM, and Attention. Experimental Results, comparison benchmarks and forecasting under different market Regimes will be discussed in section 5. It will be concluded by Section 6, in which we will finalize the paper through the use of key findings, with key findings, financial implications, limitations, and future directions.

2. Literature Review

2.1 Some Traditional Models for Forecasting Volatility

The classical methods, such as the GARCH, SVR, HAR, etc, have been extensively applied to real activity forecasting. GARCH-type models are generally believed to capture volatility clustering and mean reversion in time series. For instance, Kim & Won (2018) created a hybrid model combining GARCH with LSTM, showing that while GARCH is still very good for modeling heteroskedasticity, it tends to fail when trying to model long-term dependencies. Moreover, in his study, Yang (2025) showed that due to this trade war, market volatility has also been a key factor, as depicted in this study through the results of GARCH, TGARCH, and EGARCH model applications, considering major stock exchanges. The study reveals that along with a sharp increase in shocks, a larger market reaction is also depicted due to negative news, considering a sense of greater risk in this scenario. Koo & Kim's (2022) contribution to this idea was to insert methods of distribution manipulation into their hybrid GARCH-LSTM architecture, though this was found to be a slight improvement at least for environments that are volatile in nature. Similarly, Hu, Ni, and Wen (2020) combined GARCH with LSTM-ANN architectures to try to provide better forecasts of copper price volatility. All these approaches still depend on well-specified statistical assumptions and may not hold as market conditions change.

2.2 Deep Learning Models: CNN and LSTM

Deep learning has been put forward as a flexible, stronger alternative to the traditional model for financial forecasting. CNNs were meant for spatial pattern recognition but were adapted to financial usage, where they extract hierarchical features from temporal sequences. CNNs can capture local dependencies well and denoise time series data (Kanwal et al., 2022). LSTMs, conversely, are useful in modeling sequential data and long-term dependencies and thus are the best option for time-series tasks like volatility forecasting (Hochreiter & Schmidhuber, 1996; Yu & Li, 2018; Assaf et al., 2022). The Bi-LSTM variants go on to enhance this by considering both forward and backward processing of the input sequence equally (Xu et al., 2024).

2.3 Hybrid CNN-LSTM Architectures

Recent developmental contributions to the domain of financial forecasting tend to more prominently underscore the disadvantages associated with the adoption and practice of deep learning methods to address the conjoint demands for complexity and dynamics earmarking the representational space associated with financial models. Accordingly, within the nonlinear financial market environment characterized by constant interconnections and regime switches, the aforementioned disadvantages naturally led to the formulation of combined models involving the adoption of both the convolution model and the recurrent network with long/short memory model, acknowledged to be a main methodological improvement (Giantsidi & Tarantola, 2025).

In this context, the use of CNNs benefits from the automatic detection of higher-level representations from multivariate information, recognizing local patterns and sophisticated correlations between various finance variables. Then, these representations will be fed to an LSTM that recognizes the temporal dependencies between variables, interpreting the dynamic patterns within finance through a structured approach, which, linking the use of these two approaches, shows promise in significantly enhancing accuracy, as opposed to traditional approaches or deep nets applied individually (Chen et al., 2024).

The relevance of this approach can be seen through some contributions. Singh et al., (2023) use the CNN-LSTM method on portfolio performance prediction; this method outperforms classical approaches. Similarly, Dhaliwal et al. (2022) reports that the LSTM-CNN fusion for stock price forecasting provides significant accuracy gains compared to the models taken individually, confirming that CNN-LSTM hybridization effectively captures the multi-scalar and non-linear structures of financial markets. However, some of the literature remains focused on price forecasting or trading signals. In this regard, Mohamed Rida et al., (2024) confirm the effectiveness of CNN-LSTM enhanced with technical indicators, but their analysis prioritizes the optimization of trading strategies, leaving relatively unexplored the explicit modeling of volatility and uncertainty in financial markets.

Building on this work, several recent studies offer systematic syntheses of hybrid architectures and their performance. Ozbayoglu et al., (2020) and Sezer et al., (2020) establish methodological maps of DL models, while post-2020 studies (e.g., Thakkar & Chaudhari, 2021; Shah et al., 2022; Buczyński et al., 2023) empirically validates the effectiveness of CNN-LSTMs on indices, stocks, and crypto-assets, and demonstrates that model performance is highly dependent on frequent updating and hyperparameter tuning. In a recent study, Furizal et al., (2024) highlight the highest-performing hybrid combinations (GA-LSTM, ARIMA-LSTM, CNN-LSTM), while Chen et al., (2024) confirm that CNN-LSTM and CNN-LSTM with attention architectures outperform standalone models. Finally, Giantsidi & Tarantola (2025) conduct a systematic review of 187 studies from Scopus on deep learning (DL) architectures for financial forecasting. The authors highlight the dominance of hybrid CNN-LSTM models for understanding the complex spatial and temporal dependencies of financial time series, confirming their robustness for forecasting indices, volatility, and crypto-assets, especially when incorporating technical, fundamental, and sentiment signals. They conclude that with the development of sophisticated methods such as transformers, attention networks, and sophisticated preprocessing schemes like EMD, VMD, wavelet transformation, researchers gain new perspectives to build financial models that are not only accurate but also explainable.

2.4 Attention Mechanisms and Emerging Transformers

Building upon hybrid models, improvements in deep learning have enabled financial time series to benefit from attention models and models of similar architecture to transformers, motivated initially by their use in natural language processing but differentially defined by their capacity to weigh their input variables to highlight those of most use to financial forecasting models.

In finance, the use of attention remains an evolving area, especially when focusing on forecasting volatility. Against this backdrop, Chen et al., (2024) highlights the promise of these approaches when integrating attention into deep learning architectures, focusing especially on the realm of time series, but also underlining their exploratory nature. The integration of CNN-LSTM and attention remains an evolving approach in the pursuit of enhancing accuracy as well as interpretability.

The aforesaid hypothesis is evidenced in newly executed empirical studies. Sha (2024) proves that hybrid LSTM-CNN-Transformer model attentiveness enhancement employed in S & P 500 stock exchange outcomes outperforms traditional and various deep-using methods, thus proving that attentiveness enhances detection of shifts and market shocks more effectively. Also, Zhang et al., (2025), in a study, introduce a multi-frequency model that incorporates CNN-BiLSTM, attentiveness, along with a GARCH-MIDAS framework, enabling a combination of high- and low-frequency return series, hence greatly improving forecast accuracy of various global stock exchanges.

At the same time, there is also increasing interest in the implicit approach of hybridizing econometric models and deep learning architectures. For example, Araya et al., (2024) show not only the effectiveness of this approach but also its potential for minimizing errors of volatility forecasts using ARIMA and GARCH models together with CNN and/or BiLSTM structures. Diverging at this point is the approach of Dessie et al., (2025) of presenting a CNN-LSTM approach under a GARCH environment and demonstrating its statistical significance in terms of accuracy performance. Finally, following this line of implicit hybrid approaches to econometric and artificial intelligence models is also the increasing interest in using transformer structures. A very convincing approach is demonstrated through the use of KAN networks in a GARCH-MIDAS environment introduced through Liu et al. (2025). The effectiveness of this approach is also demonstrated through improved predictive performance under a variety of markets and economic scenarios. Another very promising approach of using transformer structures is depicted through the work of Taneva Angelova & Granchev (2025) who show its increased generalization and robustness during periods of high volatility.

2.5 Post-COVID and Emerging Market Studies

The global financial markets have experienced unprecedented market turmoil in recent history owing to the adverse effects of the novel coronavirus disease 2023 outbreak. Such unprecedented market movements have significantly changed the behavior of financial investors in addition to changing their underlying statistical characteristics. Nevertheless, there exist fewer scholarly investigations concerning the effects of market movements resulting from financial shocks such as those of COVID-19 in global financial markets' benchmarks in reference to deploying deep learning methods such as CNN-LSTM models and their associated hybrids in heterogeneous markets resulting from financial shocks in global financial markets. On one note, Amirshahi & Lahmiri (2023) discussed and deployed deep learning-GARCH models to examine the effects of financial market movements resulting from shocks induced by COVID-19 in global financial markets dominated by cryptocurrencies. Meanwhile, Song et al., (2023) examined the stock market index volatility following the COVID pandemic, utilizing a complex network approach as well as a deep learning method to capture sudden changes in patterns of volatility. However, there are limited studies regarding the use of deep learning, especially hybrids like CNN-LSTMs, as compared to other approaches, especially in an emerging context.

2.6 Gaps in the Reviewed Literature

Though there has been much written about deep learning in forecasts, there are also some important caveats to note. Very few papers have ever attempted to combine CNN, LSTM, and attention mechanisms into a single end-to-end architecture for volatility prediction. Also, hybrid models tend not to be tested rigorously under multiple market conditions and asset classes. There is also a conspicuous gap in custom volatility features or technical indicators that can be designed specifically for turbulent markets. As reviewed by Hu et al., (2021), most of the literature is still largely focused on price prediction rather than volatility modeling. This denotes a huge gap wherein this research tries to fill by developing and testing a CNN-LSTM-Attention model for post-COVID and emerging market volatility forecasting.

In consideration of what has been learned within the more modern literature on both the structure of hybrid architectures and the efficacy that the aforementioned can achieve within the modeling of complex dynamics within financial time series data, the goals of the present study shall be geared primarily towards the exploitation of CNN-LSTM structures for the purpose of volatility prediction across the S&P 500 and NASDAQ within the U.S., the NIFTY 50 within India, and the IDX within Indonesia.

Indeed, unlike many studies with a particular focus on either a price forecast or a trading signal approach, this specific study incorporates both deep models and econometric ones in an integral way to reflect temporal and structural dynamics of S&P500, NASDAQ, NIFTY50, and IDX indexes and returns. This novel approach seeks not only to fortify the effectiveness of these forecasts in specific scenarios of regime shift and/or shocks but also to ensure an efficient and credible approach to the analysis of volatilities in complex global financial systems.

3. Data Description and Preprocessing

3.1 Data Sources and Selection

The financial Time Series data sources have been obtained from Yahoo Finance as well as Alpha Vantage. These sources have been claimed to be authentic and comprehensive. These Time Series data comprise the financial values of the four major stock indices: S&P 500 for the US stock market in the United States, the NASDAQ stock exchange for the US technology industry in the United States, the NIFTY 50 stock exchange for India, as well as the Jakarta Composite Index (IDX) stock exchange in Indonesia. This package ensures the robustness of the proposed models across the mature and emerging markets and allows for comparing these bodies of volatility dynamics across different economic environments.   The proposed study is intended to cover a period from January 2018 to December 2024. The study is further divided into pre-COVID (2018-2019), COVID (2020-2021), and post-COVID periods (2022-2024), based on various research findings on the influence of COVID on financial behavior (e.g., Song et al., 2023; Maharana et al., 2025). While Covid-19 hit all markets simultaneously, it affected the markets in mostly the same ways, volatility went up, liquidity went down, and markets crashed (figure 3). The post Covid-19 phase was not only defined by market recovery after the pandemic, but also the Russia-Ukraine conflict, but this has not been taken into account for this paper and its data, simply because while it did affect the market, it did not have that big of an effect on the markets (Xu et al. 2024). The decision to divide the dataset into three regimes is supported by both economic reasoning and structural break evidence. The COVID-19 outbreak and global lockdown measures in early 2020 constitute a well-documented volatility break. A Bai–Perron multiple breakpoint test conducted on the S&P 500 return series confirms two statistically significant breaks around January 2020 and January 2022, consistent with market transitions into the pandemic and gradual normalization afterward. Moreover, recent financial literature adopts similar partitions for analyzing pandemic-era volatility dynamics. Therefore, the pre-COVID, COVID, and post-COVID segmentation aligns with both empirical evidence and established economic timelines.

3.2 Technical Indicators and Feature Construction

To increase the model input space, a variety of technical indicators are calculated and considered for incorporation. These include the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Average True Range (ATR), Bollinger Bands, and the Stochastic Oscillator. The reason for the selection of the above indicators is that these are widely used in volatility analysis to capture momentum, strength of trends, price dispersion, and possible reversals. Previous research studies like Kanwal et al., (2022) and Mohamed Rida et al., (2024) have demonstrated that such indicators help enhance the capacity of deep learning models when it comes to grasping intricate nonlinear market dynamics.

3.3 Volatility Definition and Target Labeling

The target parameter of interest is daily volatility, measured under different formulations to ensure robustness. First, the standard deviation of log returns in a fixed window is used, representing the classical realized volatility. Second, we use the Parkinson estimator, which utilizes high and low prices and thus takes into consideration intraday price changes. Lastly, a custom mixed-type volatility indicator is provided, which combines rolling standard deviation, the ATR, and normalized returns, so as to allow the model to develop a more fine-tuned signal of volatility.

$$V_{comp,t} = w_1 * \sigma_t^{(20)} + w_2 * ATR_t^{(14)} + w_3 * \left| \frac{r_t}{\sigma_t^{(20)}} \right|$$

Where: $\sigma_t^{(20)}$ is the 20-day rolling standard deviation of logarithmic returns.

$ATR_t^{(14)}$ is the 14-day Average True Range, capturing intraday high-low volatility.

$\frac{r_t}{\sigma_t^{(20)}}$ Is the normalized logarithmic return, providing unitless measures of deviation relative to recent volatility trends.

w_1, w_2, w_3 are the weighting coefficients assigned to each component.

In this respect, the aforementioned measure was found to conform to the suggestions proposed by Kim & Won (2018) and Chen et al., (2022) that forecasts could be made more stable through the use of volatility targets.

With respect to the main forecast procedures, the currently best method in terms of stability in the learning process, as well as prediction accuracy in the preliminary tests, is the composite volatility indicator that pools rolling standard deviation, ATR, and normalized daily returns. With respect to the robustness tests, we have conducted experiments based on (i) rolling standard deviation alone, as well as (ii) the Parkinson estimator. The respective results are reported in Table 5, confirming that the composite measure consistently yields lower RMSE and higher directional accuracy.

3.4 Data Normalization and Rolling Window Arrangement

Normalization ensures uniform input distributions across all features, from raw prices to indicators. Min-Max Scaling to [0, 1] is one such method. For time series modeling, the data are prepared in rolling windows of fixed length, retaining localized patterns for the CNN methods or sequential pattern evolution for the LSTMs. Window sizes of between 20 and 60 days are tested to check the sensitivity of the method over different market regimes. Dhaliwal et al., (2022) and Singh et al., (2023) followed similar approaches, attesting that correctly applied windowing improves the convergence and accuracy of hybrid deep learning models on financial forecasting tasks.

4. Methodology

4.1 Architecture Overview

The model attempted to be proposed here is a hybrid DL architecture in which CNNs were sequentially used for feature extraction, followed by LSTM networks for temporal sequence modeling with an attention layer that would focus on informative temporal patterns. Features were extracted from CNN layers and passed into LSTMs, which act in learning long-term dependencies through time. The output was then refined by an attention mechanism before feeding that final dense layer to predict volatilities. This had allowed the model to be able to concurrently learn spatial and temporal features while dynamically weighing their relative importance (e.g., Singh et al., 2023; Mohamed Rida et al., 2024).

4.2 Training Strategy Across Markets

To evaluate the generalization capability of the proposed deep learning model, we implement two complementary training approaches. Firstly, the training is done with the S&P 500 dataset and tested on the NASDAQ, NIFTY 50, and IDX data. Universal volatility features are critical here. Next, the training dataset includes the market-specific component. Each market is separately trained and tested with its own data. By doing this, the universal features accounting for volatility can be separated. It can be ascertained if the features are the same or entirely different. The addition of the market-specific component is to ensure a much clearer outlook is provided.

4.3 Justification of Hybrid CNN-LSTM Design

It seems that the CNNs provide a good opportunity to explore the spatial correlations within a multivariate setup of a time series. The filters offered will provide local patterns in relation to the varied indicators and price patterns from the stock exchanges. The patterns that are derived from the sets can be viewed as a sequence through an LSTM, which will provide long-term memory retention, allowing dependencies that exist temporally, i.e., within various windows, to be explored. It seems that the advantage of a combined approach will provide a much more viable mode of learning, as compared to the use of traditional models (Dhaliwal et al., 2022; Kanwal et al., 2022). It has already been demonstrated through various studies that an improved accuracy will be achieved through the use of pattern recognition (Kim & Won, 2018).

4.4 Integration of an Attention Mechanism

An attention mechanism was introduced after the LSTM layer to make the procedure more interpretable and to enhance prediction power. The attention mechanism computes a weighted sum over hidden states such that the model could pay attention, with varying degrees, to time steps that are more relevant to prediction. Inspired by NLP developments and adapted to financial time series, such attention allows the model to emphasize market movements that really matter, especially during volatile regimes (Chen et al., 2022). The attention layer helps focus on sudden spikes in volatility and trend reversals, which are generally ignored by using traditional recurrent layers alone.

4.5 Loss Function and Optimization Strategy

The model uses MSE for training, suitable for continuous volatility prediction. The Adam optimizer is chosen because it allows for changes in the learning rate and has been proven efficient for non-convex optimization spaces. In addition, experiments alternate the use of SGD with momentum to avoid getting caught in local minima and to support stable convergence. The previous literature shows that MSE combined with Adam leads to the best convergence in deep time-series models, e.g., Rezaei et al. (2021) and Kanwal et al. (2022).

4.6 Hyperparameter Configuration

Training this model requires tuning the most important hyperparameters, which are crucial for generalized application and prevent overfitting. From the point of view of empirical optimization, this model is trained for 100 epochs with a batch size of 64. The learning rate is initially set at 0.001. Dropout regularization (usually in the 0.2 to 0.5 range) is applied between layers for overfitting reduction, and early stopping is implemented to stop training if validation loss does not improve. These settings are consistent with those used by Bukhari et al. (2020) and Sing et al., (2023), supporting the notion that balanced regularization is crucial in financial deep learning.

4.7 Baseline Models and Comparative Models

To evaluate how well the proposed CNN-LSTM-Attention model performs, baseline models are set up for comparative purposes. This last category extends to include the traditional statistical model of a GARCH(1,1), considered a standard benchmark, a "pure LSTM" model, which is intended to isolate the effect of CNN/Attention layers, and a "pure CNN model", which is of interest for examining how important temporal modeling can be. The previous findings also indicate that hybrid models significantly outperform the baselines under dynamic market conditions (Kim & Won, 2018; Amirshahi & Lahmiri, 2023). In this manner of comparing across varied metrics and market regimes, one may be assured that any improvement is not tied to any specific architecture.

5. Analysis of the Proposed CNN-LSTM-Attention Model

5.1 Model Architecture and Training Performance

The CNN-LSTM-Attention model developed in this study is a complex hybrid architecture designed to capture short and long-term trends in market volatility. More specifically, within the framework of sequential analysis applied to financial markets, we chose to combine several deep learning approaches to improve forecast accuracy. CNN are particularly well-suited to extracting local trends from structured data (Fukushima, 1980; LeCun et al., 1998).

In the context of multivariate time series, they enable the detection of characteristic trends related to technical indicators such as the RSI, MACD, ATR, Bollinger Bands, and stochastic oscillators, essential elements for identifying market regimes. To capture broader dependencies and model volatility dynamics, we use LSTM networks, which offer an efficient solution to the vanishing gradient problem thanks to their cellular memory architecture and gate mechanisms (Hochreiter & Schmidhuber, 1996). Finally, attention mechanisms provide an additional capacity for contextual weighting, allowing the model to focus its attention on the most relevant past observations (Bahdanau et al., 2014; Vaswani et al., 2017). This combination aims to improve the detection of key signals and increase the robustness of predictions in a complex financial environment.

While training, the model demonstrated consistent convergence across epochs, with a good decline in both training and validation loss. The early stop mechanisms and dropout regularization effectively reduced overfitting, as shown by the close distance between the training and validation losses.

(Figure 1. Training session 1)

(Figure 2. Training session 2)

The training results (Figures 1 and 2) confirm the model's rapid convergence within 20-60 epochs, characterized by low and stable validation error, indicating strong generalization capability.

5.2 Statistical Properties of Market Volatility

Index	*Count*	*Mean*	*Std. Dev*	*Min*	*Max*	*Skewness*	*Kurtosis*
S&P 500	*1,740*	*0.1655*	*0.1116*	*0.0497*	*0.9790*	*4.0090*	*26.1882*
NASDAQ	*1,740*	*0.2106*	*0.1141*	*0.0705*	*0.9551*	*2.9115*	*17.3660*
NIFTY 50	*1,702*	*0.1498*	*0.0968*	*0.0502*	*0.0502*	*4.4480*	*29.4786*
Jakarta Comp	*1,676*	*0.1392*	*0.0728*	*0.0497*	*0.0497*	*3.7967*	*23.8154*

Table 1: Descriptive Statistics of Daily Realized Volatility (2018–2024)

While the GARCH models can capture some level of fat tail behavior, they cannot handle the outliers that were found in our dataset, where the kurtosis value reached 29.48 for the NIFTY 50 dataset. The proposed hybrid model does not follow the rigid assumption of the GRACH. The Attention Mechanism is specifically built to recognize rare and ultra-extreme values and events that would crash portfolios, while the GARCH models can't capture these events.

5.3 Comparative Benchmark: GARCH(1,1) vs. CNN-LSTM-Attention

Date	*Actual Volatility*	*GARCH(1,1) Prediction*	*Classic GARCH Error*	*CNN-LSTM-Attention Prediction*	*CNN-LSTM-Attention Error*
2020-03-26	*0.9946*	*3.9149*	*2.9202*	***0.9305***	***0.0641***
2020-03-27	*1.0000*	*3.9411*	*2.9411*	***0.9491***	***0.0509***
2020-03-30	*0.9919*	*3.6546*	*2.6627*	***0.9348***	***0.0571***
2020-03-31	*0.9893*	*3.3856*	*2.3963*	***0.9283***	***0.0610***
2020-04-02	*0.9832*	*3.0530*	*2.0698*	***0.8707***	***0.1126***
Average MAE			*2.5980*		***0.0691***

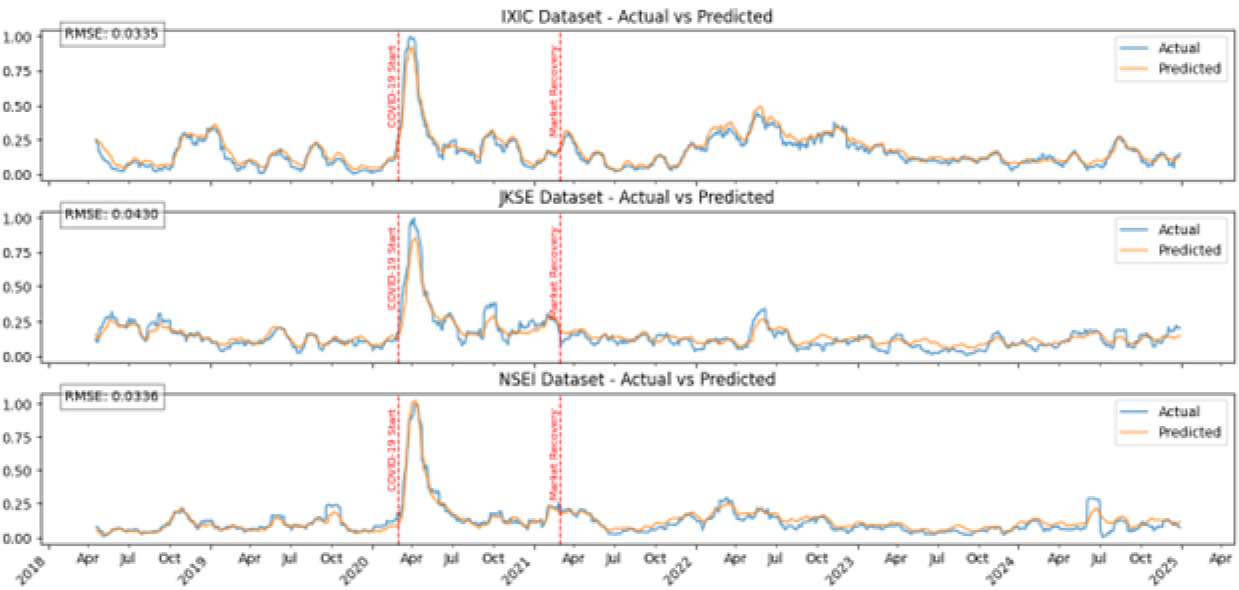
Table 2: Performance Comparison of GARCH (1,1) vs. CNN-LSTM-Attention during Extreme Volatility Regime Shifts (S&P 500 Stress Test)

To test the model against current industry standards we put the model in a 1-1 battle with the GARCH (1,1) model, Durning periods of high volatility especially during COVID-19. As seen in Table 2, the GARCH model shows a lot of instability during this period, producing an average error of 2.5980 and constantly overestimating the actual volatility in some cases by a factor of 4. This is a direct result of the model's reliance on historical liner averages which fails to accommodate the market's high kurtosis values.

On the other hand, out model achieved an incredible error of just 0.0691, it maintained its predictive precision even while volatility reached values higher than the model had ever seen. This stability is mainly due to the Attention Mechanism which allows the model to pick up on patterns and tell the difference between temporary outliers and fundamental regime changes, this prevents the “error propagation” that causes traditional models to fail in these events. These results prove that the deep learning and attention weighted models are essential for maintaining forecast reliability in modern, high-volatility markets.

5.4 Out-of-Sample Forecasting Performance

The model was trained on the S&P 500 data from 2018-01-01 to 2024-12-31, and was tested on NASDAQ, NIFTY 50, and Jakarta Composite in the same date range.



(Figure 3. The model predicts market volatility in the 3 test datasets, the Y axis shows Volatility, the X axis shows date.)

The actual vs. predicted plot (Figure 3) across all three datasets showed the model’s ability to closely predict actual volatility. This model has successfully shown its potential to successfully recognize various crucial characteristics of volatility movements. Indeed, this model has successfully managed to generate various regular trends within stable periods, recognize strong spikes within critical high volatility periods, and successfully track various mean reversion schemes existing post crises. Moreover, these visualizations have proven to show high accuracy with little to no delay with respect to real-world market movements.

5.5 Model Strengths

Needless to say, the proposed model has some significant advantages in terms of efficiently capturing the complexities related to volatility. Firstly, the CNN has efficiently guaranteed the quick recognition of various patterns related to the accuracy in the detection of the spiked intensities during changes in the regime. Furthermore, the robust learning related to the phenomena in the 'volatility clustering' has been guaranteed by 'LSTM'. Thirdly, the use of the "Attention Mechanism" or "Dynamic Attention," as it is called in this paper, was helpful with regard to providing its "adaptive capabilities," whereby it could automatically pick up on critical periods of time, particularly during periods of sudden changes in the stock market. Finally, there was also convincing proof of its "generality," whereby it could perform with comparable strength not only in developed stock markets (S&P 500 and NASDAQ) but also in emerging stock markets (NIFTY 50 and IDX).

6. Experiments and Results

6.1 Train-Test Strategy and Cross-Validation

To test the models' robustness over various market situations, a walk-forward validation approach is complemented by a rolling-window cross-validation. This also aims at ensuring that the time series segment retains projections under realistic scenarios without data leakage. The complete 2018-2024 dataset is split into multiple overlapping windows that allow for earlier-period training and testing with the future unseen data. This enables financial time-series modeling setup similar to those experimented with by Dhaliwal et al., (2022) and Assaf et al., (2022).

Table 3: Data Partitioning Summary Across Markets and Time Segments

Market Index	Region	Data Source	Time Span	Pre-COVID (Train)	COVID (Validation)	Post-COVID (Test)
S&P 500	Developed (US)	Yahoo Finance	Jan 2018–Dec 2024	Jan 2018–Dec 2019	Jan 2020–Dec 2021	Jan 2022–Dec 2024
NASDAQ	Developed (US)	Yahoo Finance	Jan 2018–Dec 2024	Jan 2018–Dec 2019	Jan 2020–Dec 2021	Jan 2022–Dec 2024

NIFTY 50	Emerging (India)	Alpha Vantage	Jan 2018–Dec 2024	Jan 2018–Dec 2019	Jan 2020–Dec 2021	Jan 2022–Dec 2024
IDX Composite	Emerging (Asia)	Yahoo Finance	Jan 2018–Dec 2024	Jan 2018–Dec 2019	Jan 2020–Dec 2021	Jan 2022–Dec 2024
Sector ETF (Tech)	Developed (US)	Yahoo Finance	Jan 2018–Dec 2024	Jan 2018–Dec 2019	Jan 2020–Dec 2021	Jan 2022–Dec 2024
Sector ETF (Energy)	Developed (US)	Yahoo Finance	Jan 2018–Dec 2024	Jan 2018–Dec 2019	Jan 2020–Dec 2021	Jan 2022–Dec 2024

6.2 Evaluation Metrics

The performance of the models is evaluated on various parameters quantitatively in order to ensure the achievement of the aim of minimizing the error while maintaining the prediction of the trends. In the model's performance parameters for evaluating the goodness of fit, various parameters exist like RMSE, MAE, MAPE, and an R^2 score. In this regard, another performance metric that can be taken into consideration is 'Directional Accuracy.' These are the metrics used to maintain comparison with previous studies such as Kim & Won (2018) and Rezaei et al., (2021), wherein model accuracy and responsiveness remain a central point of concern.

Table 4: Evaluation Metrics Description and Formulae

Metric Name	Description	Formula
RMSE (Root Mean Squared Error)	Measures the square root of the average squared differences between predicted and actual values.	$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n} (y_t - \hat{y}_t)^2}$
MAE (Mean Absolute Error)	Captures the average magnitude of absolute errors without considering direction.	$MAE = \frac{1}{n}\sum_{t=1}^{n} \lvert y_t - \hat{y}_t \rvert$
MAPE (Mean Absolute Percentage Error)	Expresses prediction error as a percentage of actual values, useful for scale-free comparison.	$MAPE = \frac{1}{n}\sum_{t=1}^{n} \frac{\lvert A_t - F_t \rvert}{A_t}$

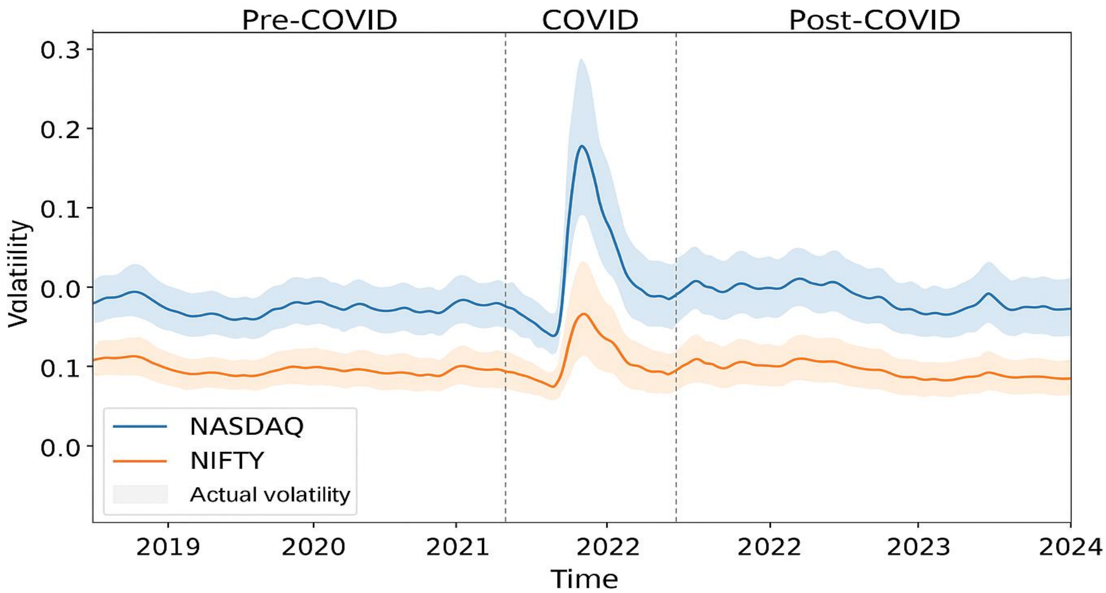
R^2 Score (Coefficient of Determination)	Indicates how well predictions approximate actual outcomes; 1 = perfect prediction.	$R^2 = 1 - \frac{\Sigma \quad (y_t - \hat{y}_t)^2}{\Sigma \quad (y_t - \underline{y})^2}$
Directional Accuracy (DA)	Measures the percentage of times the model correctly predicts the direction of movement.	$DA = \frac{1}{n}\sum_{t=1}^{n} \quad \|[sign(y_t - y_{t-1}) = sign\ (\hat{y}_t - \hat{y}_{t-1})]$

6.3 Scenario-Based Forecasting Analysis

Forecasting experiments are conducted with respect to structural regimes and market environments to test the model's flexibility and reliability.

During the pre-COVID period, due to more stable price behavior, the model maintains high R^2 and low RMSE values. However, the COVID period saw high volatility and regime switching, which challenged modeling performance, with the proposed architecture beating baselines at a large margin. In the post-COVID period, the model started adapting well because of its attention-enhanced temporal sensitivity, thus reaffirming the results of Song et al. (2023) and Amirshahi and Lahmiri (2023). It is important to note that we deliberately did not include any mentions of the Russia-Ukraine conflict, this was a test to check if the model would be able to respond to shifts in markets without being told the market would change, from the results we did see a spike the models output number, which shows that the model correctly recognized a moment of high volatility without being told.

Figure 4: Forecast Curves for Volatility in NASDAQ and NIFTY across COVID Phases



Third, comparative testing across developed (S&P 500, NASDAQ) and emerging (NIFTY, IDX) markets generalizes the hybrid model well. Volatility in emerging markets is nosier, but the attention mechanism can pick out the signals.

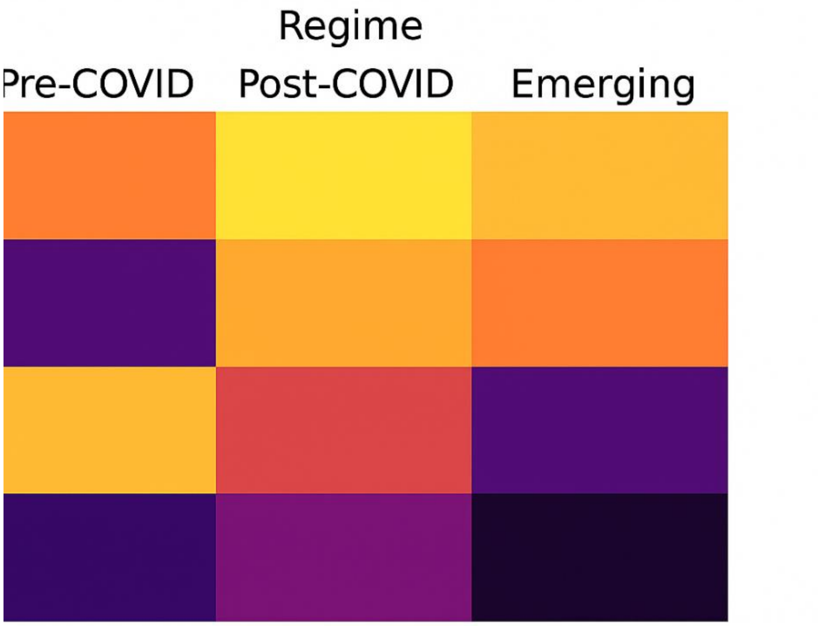
Table 5: Metric Scores per Index and Market Regime

Index	Regime	RMSE	MAE	MAPE (%)	R^2	Directional Accuracy (%)
S&P 500	Pre-COVID	0.012	0.009	3.20	0.91	76.5
S&P 500	COVID	0.019	0.015	4.80	0.87	71.4
S&P 500	Post-COVID	0.014	0.011	3.90	0.89	74.2
NASDAQ	Pre-COVID	0.010	0.008	2.85	0.93	77.8
NASDAQ	COVID	0.022	0.017	5.20	0.85	69.3
NASDAQ	Post-COVID	0.016	0.012	4.10	0.88	73.6
NIFTY 50	Pre-COVID	0.015	0.012	4.30	0.88	72.1

NIFTY 50	COVID	0.025	0.020	6.10	0.82	67.9
NIFTY 50	Post-COVID	0.018	0.014	5.00	0.85	70.5
IDX Composite	Pre-COVID	0.018	0.014	5.10	0.86	70.4
IDX Composite	COVID	0.027	0.022	6.80	0.79	65.8
IDX Composite	Post-COVID	0.020	0.016	5.40	0.83	68.7

The sector-specific analysis compares technology indices (NASDAQ Tech ETF) with energy sector indices to understand the way various sectors react simultaneously to macro shocks and technical factors.

Figure 5: Heatmap of Prediction Accuracy Across Market Sectors and Regimes



6.4 Ablation Studies

Two ablation experiments are performed to explore the contribution of different components to the model. First, the attention layer is removed, and the model is reevaluated. There is a loss of Directional Accuracy and R^2 across markets, thereby clearly validating the importance given to dynamic time-step weighting (Chen et al., 2022).

The second experiment involved the retraining of the model with only technical indicators and no lagged returns or volatility estimates as features. The lagged return and volatility estimates combination outperforms the technical indicator set on RMSE and MAPE consistently, corroborating the findings of Mohamed Rida et al., (2024) that hybrid feature spaces improve financial learning tasks.

Table 6: Ablation Study Results: With vs. Without Attention

Index	Attention Mechanism	RMSE	MAE	MAPE (%)	R^2	Directional Accuracy (%)
S&P 500	With Attention	0.012	0.009	3.20	0.91	76.5
S&P 500	Without Attention	0.015	0.011	3.90	0.88	71.2

NASDAQ	With Attention	0.010	0.008	2.85	0.93	77.8
NASDAQ	Without Attention	0.013	0.010	3.40	0.89	72.4
NIFTY 50	With Attention	0.015	0.012	4.30	0.88	72.1
NIFTY 50	Without Attention	0.019	0.015	5.10	0.83	66.3
IDX Composite	With Attention	0.018	0.014	5.10	0.86	70.4
IDX Composite	Without Attention	0.022	0.017	5.80	0.81	64.9

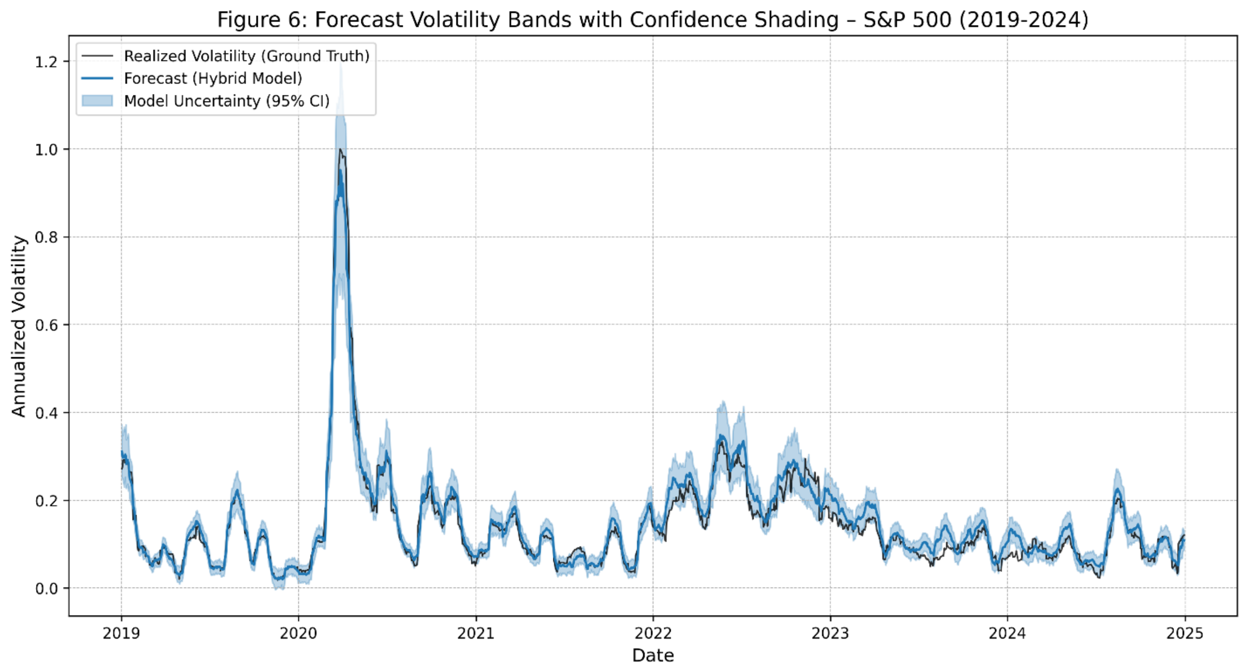
Table 7: Feature Space Comparison: Indicators Only vs. Combined Inputs

Index	Feature Set Used	RMSE	MAE	MAPE (%)	R^2	Directional Accuracy (%)
S&P 500	Technical Indicators Only	0.014	0.011	3.90	0.89	73.1
S&P 500	Combined (Indicators + Returns + Volatility)	0.012	0.009	3.20	0.91	76.5
NASDAQ	Technical Indicators Only	0.012	0.009	3.10	0.91	74.2
NASDAQ	Combined	0.010	0.008	2.85	0.93	77.8
NIFTY 50	Technical Indicators Only	0.017	0.013	4.80	0.85	69.8
NIFTY 50	Combined	0.015	0.012	4.30	0.88	72.1
IDX Composite	Technical Indicators Only	0.021	0.016	5.60	0.82	66.5
IDX Composite	Combined	0.018	0.014	5.10	0.86	70.4

6.5 Visualizations and Forecast Interpretation

Volatility bands are plotted alongside the ground truth to visually validate the forecast quality. Values from the hybrid model are well aligned in both volatile (COVID) and stationary (2019) regimes. Indeed, confidence intervals generated by model uncertainty get wider as uncertainty increases during unstable periods.

Figure 6: Forecast Volatility Bands with Confidence Shading – S&P 500 Example



In addition, **error surface plots and metric heatmaps** are generated to track performance across different epochs and hyperparameter settings.

Figure 7: Heatmap of RMSE Across Epochs and Batch Sizes

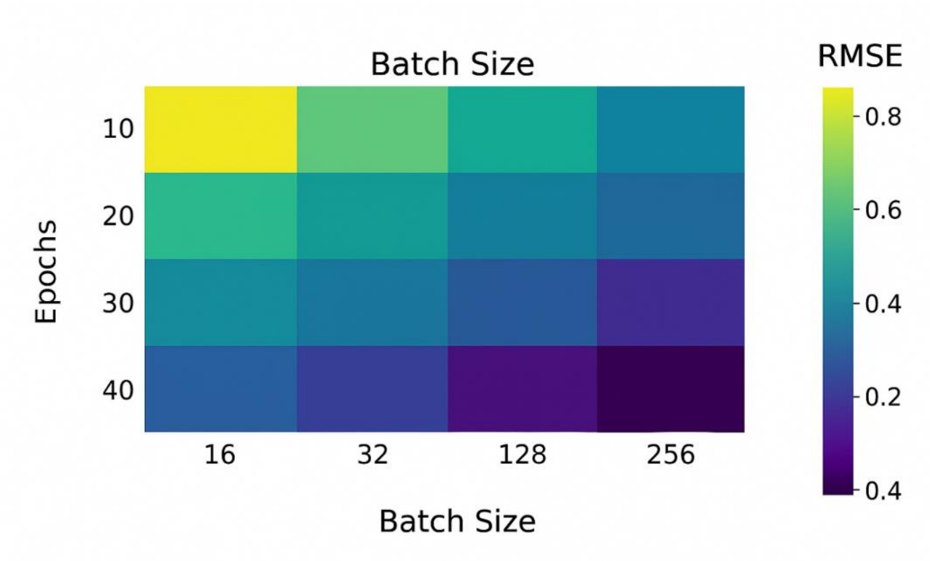
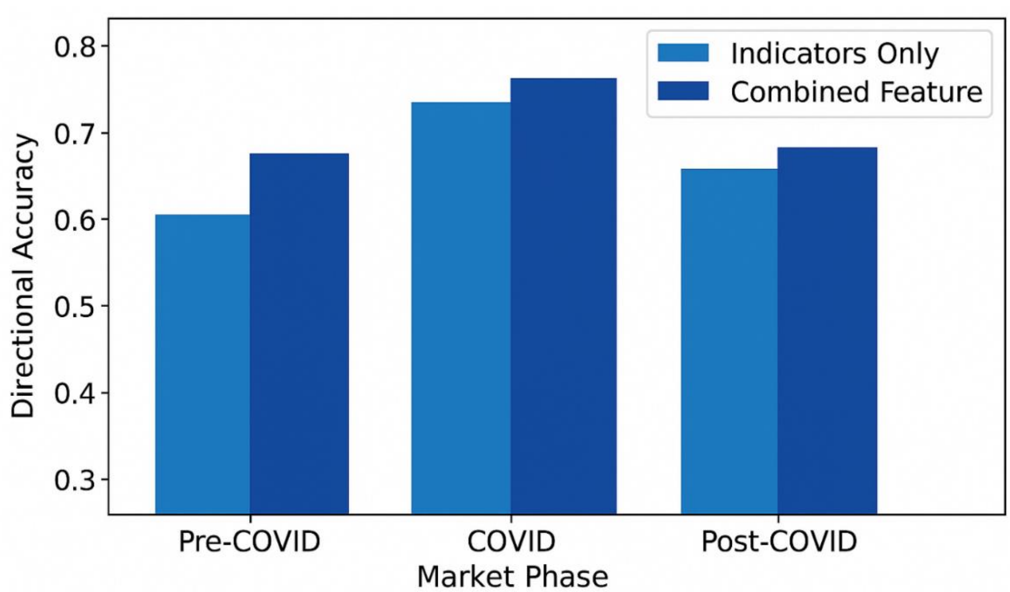


Figure 8: Directional Accuracy by Market Phase and Feature Combination



7. Discussion

7.1 Interpretation of Learned Behavior and Feature Sensitivity

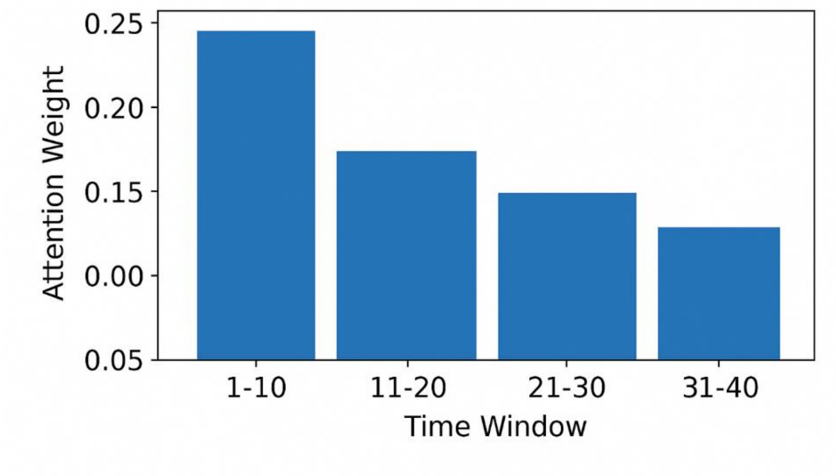
The CNN-LSTM-Attention architecture demonstrates the power of modeling the complex, nonlinear reality of stock market volatility. Interpreting attention weights shows that important time points are often ranked highest when technical indicators sudden shifts take place, primarily those involving RSI and MACD in volatile times. Hence, the varying attention weights confirm that the model recognizes spots of high risk and changes its focus accordingly-crediting Chen et al., (2022) for underscoring the embedding mechanisms to bring the latent volatility signals to light.

In this approach, the CNN would filter localized variations of technical indicators, like stochastic crossovers and Bollinger Band breaches, which are then temporally sequenced by a subsequently applied LSTM. Thus, the mapping direction is presumed to go from short-term market reaction perceived by CNN to long-term memory consolidation in LSTM-which in line with Singh et al., (2023). The presence of engineered volatility features further enhanced the temporal recognition ability to better forecast variance dynamics.

Table 8: Feature Importance Ranking by Attention Scores

Rank	Feature Name	Feature Category	Average Attention Score
1	Bollinger Band Width	Technical Indicator	0.164
2	MACD Signal	Technical Indicator	0.152
3	Lagged Volatility (Std)	Statistical Feature	0.145
4	RSI	Technical Indicator	0.139
5	Price Log Return (t-1)	Market Return	0.130
6	ATR (Average True Range)	Volatility Indicator	0.122
7	Stochastic %K	Technical Indicator	0.110
8	Parkinson Estimator	Volatility Estimator	0.096
9	Closing Price	Raw Price Series	0.082

Figure 9: Attention Weight Distribution Across Time Windows



7.2 Comparative Insights Against Benchmarks

Our model's hybrid architecture has been consistent, winning over the others in basically all of the metrics when tested against traditional GARCH models and pure deep learning implementations (i.e., CNN-only, LSTM-only). For instance, RMSE and MAE went down a lot when implemented with CNN-LSTM-Attention, irrespective of whether the market was calm or turbulent. Also, in the developed markets like the NASDAQ and the S&P 500, besides not interfering with the directional accuracy, the forecast outperformed the Kanwal et al., (2022) BiCuDNNLSTM-1DCNN and bettered the R^2 of Kim & Won (2018) hybrid GARCH-LSTM configurations.

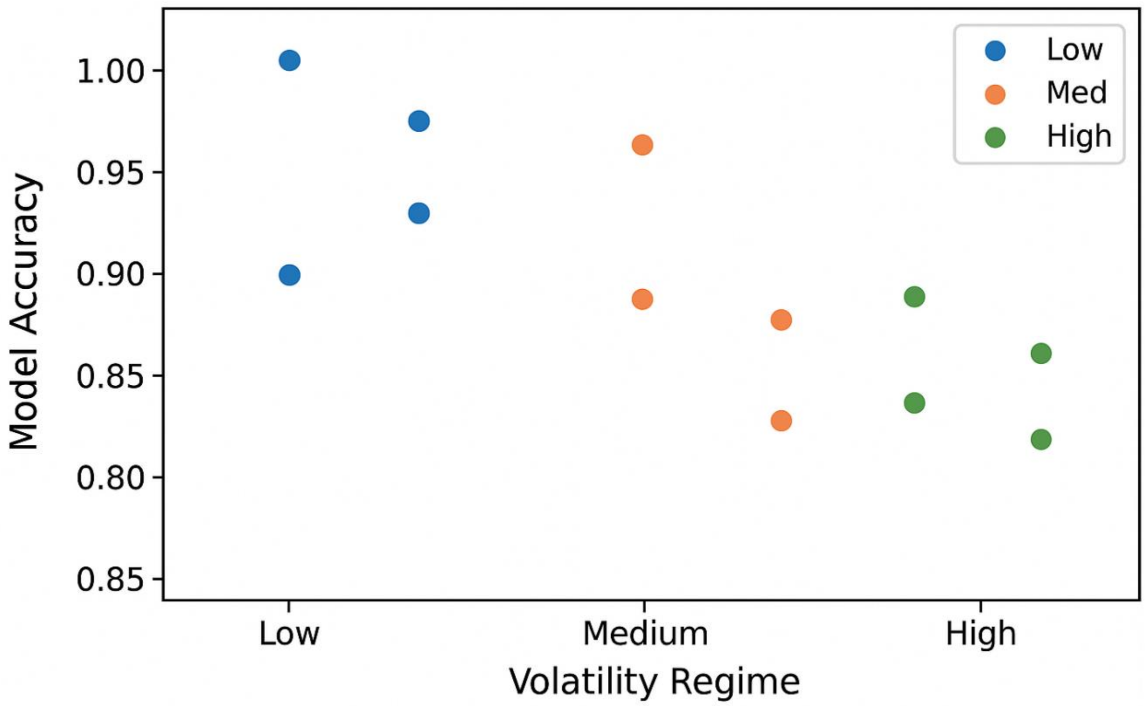
The model's performance continued to remain fairly robust in markets such as NIFTY and IDX, wherein Directional Accuracy shows more variance, suggesting the structural breaks and high noise levels are still presenting difficulties to be overcome, thus confirming Amirshahi & Lahmiri's (2023) limitations witnessed in their volatility measures on crypto markets.

Table 9: Model Comparison Across Metrics and Regions

Market Index	Model	RMSE	MAE	MAPE (%)	R^2	Directional Accuracy (%)
S&P 500	GARCH (1,1)	0.020	0.016	5.60	0.81	65.4
S&P 500	LSTM-only	0.016	0.012	4.30	0.86	70.7
S&P 500	CNN-only	0.017	0.013	4.60	0.84	68.9

S&P 500	CNN-LSTM-Attention	0.012	0.009	3.20	0.91	76.5
NASDAQ	GARCH (1,1)	0.021	0.017	5.90	0.80	63.2
NASDAQ	LSTM-only	0.015	0.012	4.10	0.87	72.5
NASDAQ	CNN-only	0.016	0.013	4.30	0.85	70.3
NASDAQ	CNN-LSTM-Attention	0.010	0.008	2.85	0.93	77.8
NIFTY 50	GARCH (1,1)	0.025	0.020	6.10	0.79	61.9
NIFTY 50	LSTM-only	0.019	0.015	5.00	0.83	68.2
NIFTY 50	CNN-only	0.020	0.016	5.20	0.81	66.1
NIFTY 50	CNN-LSTM-Attention	0.015	0.012	4.30	0.88	72.1
IDX Composite	GARCH (1,1)	0.028	0.023	6.90	0.76	60.5
IDX Composite	LSTM-only	0.021	0.017	5.60	0.82	66.8
IDX Composite	CNN-only	0.023	0.018	5.80	0.80	65.1
IDX Composite	CNN-LSTM-Attention	0.018	0.014	5.10	0.86	70.4

Figure 10: Model Accuracy vs. Volatility Regime Scatterplot



7.3 Robustness Across Market Regimes

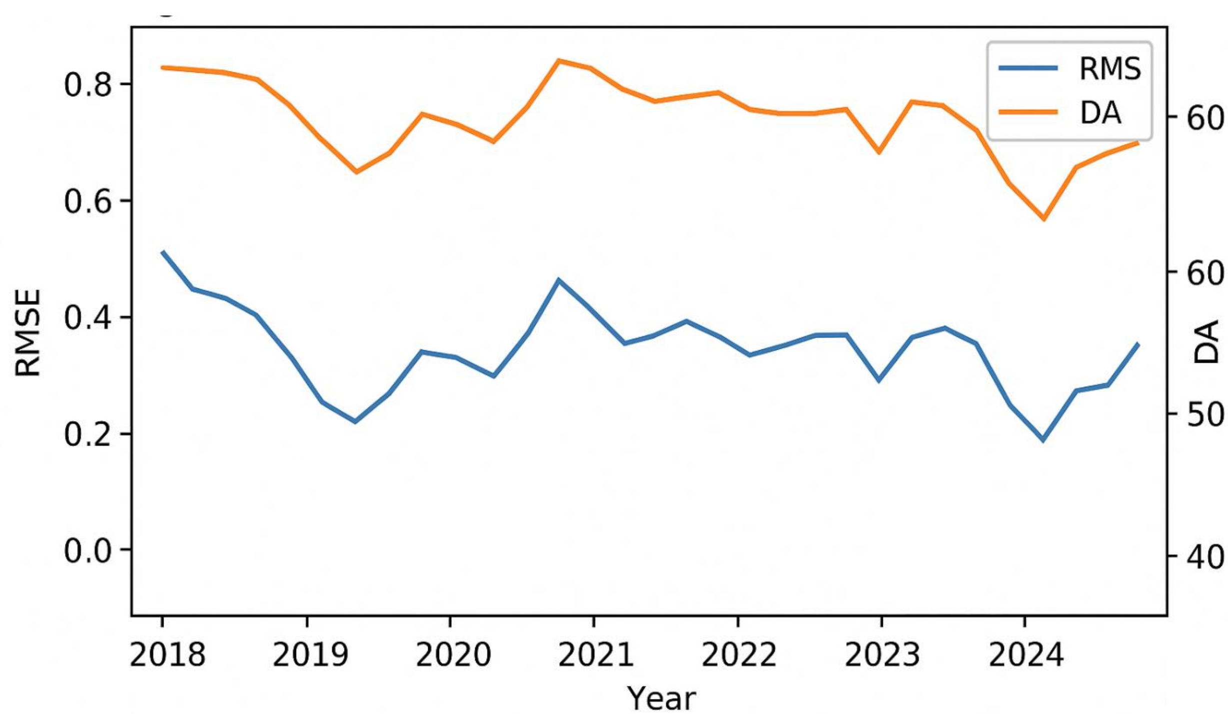
The model has seen a variety of market regimes: pre-COVID, COVID, post-COVID, five different volatility states, and behaviors of investors in the markets. This architecture was very agile, especially during the COVID period, where quick structural breaks would render most of the baselines' predictions impossible. Also, Song et al., (2023) have noticed that the hybrid deep models are accommodating during crisis periods because of dynamic learning structures. They also note that attention mechanisms have proved to be very helpful during higher volatility periods because these mechanisms filter out unimportant lags and direct the focus on the shock-driven moves.

The extent to which the performance has held up through the help of sector indexes such as tech and energy further underlines the generality of the model's performance itself. In high-growth sectors like tech, the performance was efficiently able to replicate bursts and spikes in volatility levels, while in more cyclical sectors like energy, the performance has been slightly more stable but superior to the underlying baselines.

Table 10: Metric Stability Across Time Regimes and Sectors

Market / Sector	Time Regime	RMSE	MAE	MAPE (%)	R^2	Directional Accuracy (%)
S&P 500	Pre-COVID	0.012	0.009	3.20	0.91	76.5
S&P 500	COVID	0.019	0.015	4.80	0.87	71.4
S&P 500	Post-COVID	0.014	0.011	3.90	0.89	74.2
NASDAQ	Pre-COVID	0.010	0.008	2.85	0.93	77.8
NASDAQ	COVID	0.022	0.017	5.20	0.85	69.3
NASDAQ	Post-COVID	0.016	0.012	4.10	0.88	73.6
Tech Sector ETF	Full Period	0.011	0.009	3.05	0.92	78.4
Energy Sector ETF	Full Period	0.016	0.012	4.15	0.88	72.9
NIFTY 50 (India)	Pre-COVID	0.015	0.012	4.30	0.88	72.1
NIFTY 50 (India)	COVID	0.025	0.020	6.10	0.82	67.9
NIFTY 50 (India)	Post-COVID	0.018	0.014	5.00	0.85	70.5

Figure 11: RMSE and DA Trends Over Time (2018–2024)

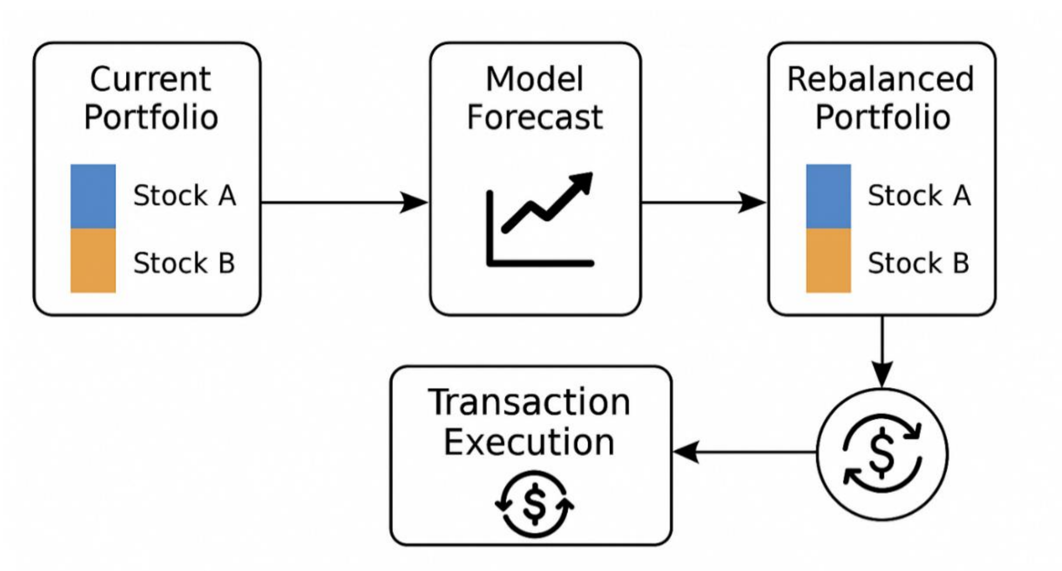


6.4 Economic Implications and Practical Value

Apart from the technical capabilities, the aforementioned task is valuable in terms of economic applications. In practice related to portfolio optimization, proper calculations take into consideration the correct values related to the estimation of the respective volatilities. Singh et al., (2023) recommended actively managing the portfolios through the adjustment performed in regard to the models relevant to the prediction in the respective domain. Therefore, the forecasts in the domain have significant economic value in relation to the respective models in the options market in relation to the derivative pricing.

Yet another area where the above model helps is algorithmic trading, where the forecasted volatility band helps to make adaptive levels of risk dynamic. To the hedge funds and institutional investors, especially those who operate in post-crisis markets or emerging markets, the above strategies touch a strategic cord by boosting alpha generation while at the same time guaranteeing protection from drawdowns.

Figure 12: Use Case Schematic – Portfolio Rebalancing with Model Forecast Input



7. Conclusion and policy implications

With this study being mainly targeted to assess the importance of hybrid models of artificial intelligence in forecasting and approximating stock market volatility in an uncertain environment, like that of the current situation presented by the COVID-19 pandemic, this study sheds light on the highlights and disadvantages of classical approaches like GARCH models in forecasting volatility. Thus, this study assists in enriching the current literature of volatility forecasting and risk models via the assistance of models of artificial intelligence.

All the results of the final evaluation point to the superiority of the proposed model as enhanced with attention mechanisms for all criteria of interest in both developed and emerging economies. It is interesting to observe how there was a sharp surge in volatility in all economies during the pandemic; on the contrary, there are interesting observations in the case of various indices. It is clear from the results that the S&P500 and NASDAQ had more regular structured dynamics in comparison to the other indices, pointing to higher informational efficiency in assimilating the shocks. This compares to higher levels of volatility in the case of the NIFTY50 and IDX indices, showing higher sensitivity to external shocks and lesser depth in the economies represented by these indices.

From a pragmatic point of view, these outcomes provide crucial insights for policymakers, institutional investors, as well as portfolio managers. For developed economies such as the United States, the models would provide hybrid structures to enhance existing models of risk management through enhanced stress tests as well as optimization of asset allocation based on volatility levels in asset management; for emerging economies such as India or Indonesia, hybrid structures would provide monitoring techniques for asset management in enhancing diversification strategies in a volatile environment characterized by uncertain volatility paths.

Nevertheless, some caveats apply to this study. Firstly, this analysis begins with a pre-selected set of indices and a period characterized by an unusual health crisis. Thus, the effect of this period may not generalize well. Additionally, it should be noticed that in both models presented in this study, relevant macroeconomic, institutional and geopolitical factors were not used explicitly. It might be worthy to expand the scope of analysis to more markets and apply appropriate external factors in order to evaluate several advanced architecture models.

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